```
int main()
{
int sockfd;
char buffer[1024];

struct sockaddr_in serverAddr;

sockfd = socket(AF_INET, SOCK_DGRAM, 0);

serverAddr.sin_family = AF_INET;
serverAddr.sin_port = htons(7564);
serverAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

strcpy(buffer,"TIME REQUEST");
sendto(sockfd,buffer,strlen(buffer)+1,0,(struct sockaddr*)&serverAddr,sizeof(serverAddr));
recvfrom(sockfd,buffer,sizeof(buffer),0,NULL,NULL);

printf("\nCurrent System Time:\n%s\n",buffer);

close(sockfd);

return 0;
}
```

output:
//server
UDP Server Running...
//client
Current System Time:
Wed Sep 9 09:53:56 2026

Exp 5:

server:

```
#include<stdio.h>
#include<stdlib.h>
```

```
#include<unistd.h>
#include<string.h>
#include<netinet/in.h>
#include<sys/wait.h>

#define PORT 7264
#define BUFFER_SIZE 1024

int main()
{
	int server_sock,client_sock;
	struct sockaddr_in server_addr,client_addr;
	char filename[BUFFER_SIZE],buffer[BUFFER_SIZE];
	server_sock=socket(AF_INET,SOCK_STREAM,0);
	server_addr.sin_family=AF_INET;
	server_addr.sin_port=htons(PORT);
	server_addr.sin_addr.s_addr=INADDR_ANY;
	bind(server_sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
	listen(server_sock,10);
	printf("File server running on PORT %d\n",PORT);
	while(1)
	{
		int client_len=sizeof(client_addr);
		client_sock=accept(server_sock,(struct sockaddr*)&client_addr,&client_len);
		pid_t pid=fork();
		if(pid<0)
		{
			perror("Fork failed\n");
			exit(0);
		}
		else if(pid==0)
		{
			close(server_sock);
			memset(filename,0,BUFFER_SIZE);
			int n=recv(client_sock,filename,sizeof(filename),0);
			if(n<0)
				close(client_sock);
			filename[n]='\0';
			FILE *fp=fopen(filename,"r");
			sprintf(buffer,"Server Process ID: %d\n",getpid());
			send(client_sock,buffer,strlen(buffer),0);
			if(fp==NULL)
			{
				sprintf(buffer,"File %s does not exists",filename);
```

```
                        send(client_sock,buffer,strlen(buffer),0);
                }
                else
                {
                        sprintf(buffer,"------File Contents---------\n");
                        send(client_sock,buffer,strlen(buffer),0);
                        while(fgets(buffer,BUFFER_SIZE-1,fp))
                        {
                                send(client_sock,buffer,strlen(buffer),0);
                        }
                        /*memset(buffer,0,BUFFER_SIZE);
                        fgets(buffer,BUFFER_SIZE-1,fp);
                        send(server_sock,buffer,strlen(buffer),0);*/
                        fclose(fp);
                }
                close(client_sock);
                exit(0);
        }
        else
        {
                close(client_sock);
                waitpid(-1,NULL,WNOHANG);
        }
    }
    close(server_sock);
    return 0;
}

client:
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<string.h>
#include<netinet/in.h>
#include<sys/wait.h>
#define PORT 7264
#define BUFFER_SIZE 1024

int main()
{
    int client_sock;
    struct sockaddr_in server_addr;
    char buffer[BUFFER_SIZE],filename[BUFFER_SIZE];
    client_sock=socket(AF_INET,SOCK_STREAM,0);
```

```
        server_addr.sin_family=AF_INET;
        server_addr.sin_port=htons(PORT);
        server_addr.sin_addr.s_addr=INADDR_ANY;
        connect(client_sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
        printf("Enter the filename:");
        scanf("%s",filename);
        //printf("%s",filename);
        send(client_sock,filename,strlen(filename),0);
        printf("Filename is sent\n");
        recv(client_sock,buffer,strlen(buffer),0);
        printf("%s",buffer);
        int n;
        /*while((n=recv(client_sock,buffer,BUFFER_SIZE,0))>0)
        {
                buffer[n]='\0';
                printf("%s\n",buffer);
        }*/
        n=recv(client_sock,buffer,BUFFER_SIZE,0);
        buffer[n]='\0';
        printf("%s\n",buffer);
        close(client_sock);
        return 0;
}
```

output:

```
//server
File server running on PORT 7264

//client
Enter the filename:exp_file.txt
Filename is sent
Server Process ID: 289618
------File Contents---------
Hello
```